CLINICAL TRIAL FAILURES · MISSED ENDPOINTS · IMMUNOLOGY & AUTOIMMUNE PHASE 2,3 · STARTS 2015–2024

JAK / TYK2: 8 of 33 completed trials missed their primary endpoint

8 completed trials across 4 sponsors and 5 molecules ran to the end and missed their primary endpoint on the sponsor's own posted result; 20 met it. Only 4 of 79 closed trials were stopped early for efficacy or safety, which is why this class is read through its completed trials rather than a stop rate. Most completed trials post nothing readable, so the count is a floor.

8 of 33
completed trials with a readable primary result missed it · 20 met it

4 / 5
sponsors / molecules behind the misses

5.1%
stopped early: 4 of 79 closed trials (95% CI 2.0%–12.3%)

62.7%
of 126 trials have closed · 47 still open or unresolved

8 of 33 **Completed and missed the primary endpoint.** Of 65 completed trials in this segment, 33 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 8 missed and 20 met, 5 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 8 of the 59 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

The completed trials that missed

Trial	Ph	Sponsor	Experimental drugs	Started	What the posted result says
NCT03934216	2	Bristol Myers Squibb	CRAVACITINIB	2019-07	Clinical Remission Response Rate at Week 12 — p 0.5935, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Odds Ratio (OR) 0.9 (95% CI 0.3 to 2.4). Method: Stratified Cochran-Mantel-Haenszel.
NCT03845517	2	Pfizer	BREPOCITINIB	2019-04	Percentage of Participants Achieving SLE Responder Index (SRI) Change of 4 (SRI-4) at Week 52 — one-sided p 0.8076, no threshold stated; 0.2 or above, so not significant at any one-sided threshold in use. Risk Difference (RD) -7.5 (95% CI -24.5 to 9.5). Method: Cochran-Mantel-Haenszel.
NCT03850483	2	Pfizer	BREPOCITINIB	2019-04	Change From Baseline in Psoriasis Area and Severity Index (PASI) Score at Week 12 — p 0.1641 against the sponsor's stated threshold of 0.05: not significant. Least square (LS) mean difference -0.6 (90% CI -1.71 to 0.43). Method: ANCOVA.
NCT03616964	3	Eli Lilly	BARICITINIB	2018-08	Percentage of Participants Achieving a Systemic Lupus Erythematosus Responder Index 4 (SRI-4) Response (4 mg Baricitinib) — p 0.711, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Odds Ratio (OR) 1.07 (95% CI 0.75 to 1.53). Method: Regression, Logistic.
NCT03134222	2	Gilead Sciences	LANRAPLENIB, FILGOTINIB	2017-05	Change in Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI) Activity Score From Baseline to Week 12 — p 0.75, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Method: Mixed-effect repeated measures model.
NCT03046056	2	Gilead Sciences	FILGOTINIB	2017-04	Percentage of Participants Who Achieved Clinical Remission at Week 24 — Risk Difference in Proportions 8.3 (90% CI -16.5 to 32.1), no p-value posted and no threshold stated: the interval is consistent with no effect under any conventional threshold, so not significant.

The 6 most recently started of 8 completed trials that missed. The diligence report lists every one, with the numbers or the sentence each verdict was read from.

How this compares

JAK / TYK2



5.1%

4/79 closed

all immunology & autoimmune trials with a resolved drug target



3.5%

30/864 closed

All immunology & autoimmune Phase 2,3



3.4%

68/1988 closed

What is in this class: 35 distinct experimental drugs across 126 trials (small molecule 126, unknown 3, antibody 2, protein 1). Most tested: FILGOTINIB (25), BARICITINIB (20), UPADACITINIB (19), TOFACITINIB (19), CRAVACITINIB (18). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 4 stops came from 3 sponsor-asset programmes across 3 sponsors; the largest (Eli Lilly) contributed 2; removing that programme's trials from both sides leaves 2.7%. Screened alongside 28 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 5 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 11.4%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	0	35	0.0%	0.0%–9.9%
2018–2020	4	35	11.4%	4.5%–26.0%
2021–2024	0	9	0.0%	0.0%–29.9%

Where the stops sit

Efficacy / futility: 2, Benefit–risk: 2.
Sponsors with most stops: Eli Lilly (2), Bristol Myers Squibb (1), AbbVie (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT03599622	2	Bristol Myers Squibb BMY	CRAVACITINIB	2022-12	Efficacy	Trial terminated because of lack of efficacy in the short term acute phase
NCT03843125	3	Eli Lilly LLY	BARICITINIB	2022-04	Benefit–risk	Study terminated due to insufficient evidence to support a positive benefit: risk profile. Safety findings were consistent with previously published OLUMIANT data
NCT03616912	3	Eli Lilly LLY	BARICITINIB	2021-11	Benefit–risk	Study terminated due to insufficient evidence to support a positive benefit: risk profile in systemic lupus erythematosus patients.
NCT03823378	2	AbbVie ABBV	ELSUBRUTINIB, UPADACITINIB	2020-09	Efficacy	Study M16-763 was terminated early as the benefit of each treatment arm from the feeder study (Study M16-063) did not provide appreciable evidence of differentiated clinical effect to warrant further long-term continuation.

Method
Denominator: ClinicalTrials.gov interventional Phase 2,3 immunology & autoimmune trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 48% of industry immunology & autoimmune trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 62.7% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.